



CAMPUS OC2 Project Kick-off Meeting

Cooperative Aggregation & Mapping for Perception in Ubiquitous Sensing

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HORIZON JU Innovation Actions | 101139048 |
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CAMPUS Team

DE HiveIntel (HIVE)

- **Role:** Semantic map enrichment and edge AI components
- **Expertise:** Semantic stream processing, reasoning, efficient AI pipelines

HU HSNLab

- **Role:** ENVELOPE integration, orchestration and networking optimization
- **Expertise:** MEC, cloud-native orchestration, B5G experimentation

HU HUN-REN TKI (Coordinator)

- **Role:** Scientific coordination, system architecture, ENVELOPE API integration
- **Expertise:** Edge/cloud systems, B5G/6G, spatio-temporal AI, ADAS services

HU NNG

- **Role:** Automotive use-case alignment and exploitation
- **Expertise:** HD Maps, navigation, AR-based ADAS

IT Scuola Superiore Sant'Anna (SSSA)

- **Role:** Network performance evaluation, orchestration strategies, geo-filtering
- **Expertise:** 5G/6G transport, AI-driven orchestration, edge computing

- Long-standing collaboration history
- Complementary expertise from AI and ADAS to networking/MEC



CAMPUS at a glance

Objective

- Enhance real-time situational awareness for connected vehicles through:
 - cooperative perception,
 - dynamic HD Map services,
 - low-latency edge-enabled processing,
 - ENVELOPE B5G network exposure APIs.



CAMPUS at a glance

Core Concepts

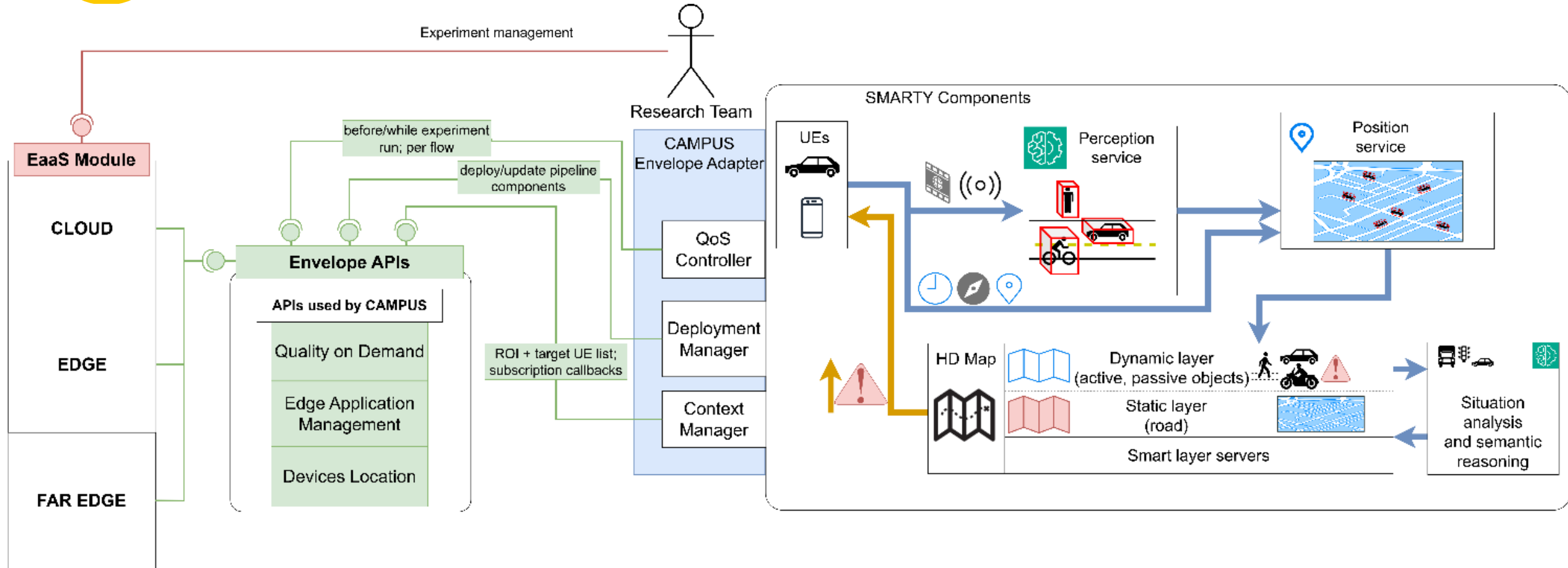
- 🚗 Multi-source Vehicle Perception
- 📱 Cooperative Sensing via Mobile UEs
- ⚡ Edge-based Spatio-Temporal Fusion
- 🗺️ Dynamic HD Map Enrichment
- 🎯 Relevance-aware Map Distribution

📶 Native usage of ENVELOPE APIs

- Quality on Demand → adaptive QoS control
- Edge Application Management → dynamic deployment/orchestration
- Device Location → context-aware filtering and delivery



SMARTY/CAMPUS architecture



CAMPUS features

-  Vehicle-based Perception
 - Camera- and LiDAR-equipped vehicles
 - Detection, tracking and short-term trajectory prediction
 - Including vulnerable road users (VRUs)
-  Cooperative Perception via Mobile UEs
 - Additional mobile sensing endpoints
 - Vehicle or portable observer UEs
 - Complementary perception streams under QoS control
 - Support for multiple perception pipelines:
 - lightweight monocular + learned depth
 - stereo / 3D object detection
 - Adaptive to different sensing capabilities of the trial environment



CAMPUS features

-  Edge-based Spatio-Temporal Fusion
 - Multi-source perception aggregation at the edge
 - Consistent, up-to-date environment representation
 - Low-latency processing close to vehicles
 - Reduced unnecessary uplink traffic toward cloud resources
-  Dynamic & Semantic HD Map Enrichment
 - NDS.Live-compatible dynamic map layers
 - Frequent updates of objects, trajectories and hazards
 - Semantic situation interpretation and conflict explanation
 - Explainability support for ADAS functions
 - Safety-relevant contextual information for ADAS functions
-  Relevance-aware Map Distribution
 - ROI-based filtering of map updates
 - Context-aware delivery to vehicles
 - Supports timely ADAS and AR-based warnings



CAMPUS-ENVELOPE integration

ENVELOPE Network Exposure APIs

- **Quality on Demand**

- Supports time-critical perception (uplink) and map-update (downlink) flows
- Predictable latency without permanent overprovisioning

- **Edge Application Management**

- Dynamic deployment and migration of perception fusion, map enrichment and semantic processing components across: Far Edge (OBU), Edge, Cloud
- Application logic remains unchanged

- **Device Location API**

- Geographic filtering & ROI-based relevance decisions
- Context-aware processing and delivery
- Devices-in-area and geofencing support
- Own http callbacks for event delivery



A large yellow graphic on the left side of the slide, consisting of a solid circle at the top left and three thick, curved lines that sweep from the bottom left towards the top right, creating a sense of motion or expansion.

KPIs/KVIs and scenarios for validation

KPIs and KVIs

	KPI/KVI Name	Description	Target Values (min – max)	
1	Map update freshness	Time from perception change at UE to availability of the updated dynamic HD Map layer at the edge/server		200 ms
2	End-to-end latency	Time from perception capture at UE to delivery of a relevant dynamic HD Map update or ADAS warning to the vehicle		300 ms
3	Relevance filtering efficiency	Ratio of delivered dynamic map updates that fall into the vehicle's region-of-interest (ROI) based on location and predicted path	85%	
4	QoS adaptation effectiveness	Ability to maintain target latency and dynamic HD Map update rates under varying network conditions using QoD (metric: requirement violation rate)		5%
5	Edge resource efficiency	Reduction in uplink bandwidth usage achieved through edge-based perception processing, compared to non-edge processing	20%	
6	Semantic interpretability (KVI)	Expert assessment of how well semantic map annotations support situation understanding and conflict explanation	Positive expert assessment Average Score: $\geq 3.5/5$	
7	Perceived safety improvement (KVI)	User and expert perception of improved situational awareness and warning usefulness	Qualitative improvement confirmed by user/expert feedback	

Planned scenarios/experiments

1 – Cooperative Perception in Occluded VRU Situations

- Multi-source perception fusion for detecting occluded VRUs
- Comparison of cooperative vs. non-cooperative ENVELOPE configurations

2 – Dynamic HD Map Updates under Adaptive Network Conditions

- Evaluation of QoD-driven adaptive QoS for map-update delivery
- Analysis of latency, freshness, jitter and delivery stability under mobility

3 – Semantic Map Enrichment and Conflict Explanation

- Edge-based semantic reasoning for intention interpretation and conflict explanation
- Validation of semantic correctness, interpretability and processing overhead

4 – Workload Orchestration between Far Edge and Edge

- Dynamic partitioning of perception and mapping workloads between OBU and edge
- Evaluation of latency, bandwidth usage and continuity during mode transitions



Planned schedule

Workplan activity	Activity description	Planned duration and effort (PM)	Expected output
Activity 1 – Preparation and integration planning	Technical alignment with the Italian ENVELOPE trial site; confirmation of ENVELOPE API interfaces; preparation of perception, fusion, HD Map and semantic components; definition of NDS.Live data layers, deployment descriptors and CI/CD setup	M1 2PM	Components ready for ENVELOPE integration
Activity 2 – Core integration and API adoption	Integration of perception and spatio-temporal fusion modules on edge resources; adaptation of HD Map generation and semantic enrichment to ENVELOPE APIs (QoD, EAM, Device Location); internal validation cycles	M1-M2 8PM	Integrated CAMPUS pipeline validated in controlled environments
Activity 3 – Initial on-site deployment	Deployment of the CAMPUS pipeline on the ENVELOPE infrastructure at the Italian trial site; configuration of QoD profiles, edge placement and location-based relevance filtering; initial test runs	M3 4PM	Operational on-site deployment
Activity 4 – Iterative experimentation and tuning	Repeated execution of experiment scenarios under varying network and mobility conditions; iterative tuning of QoD and deployment strategies, supported by remote experimentation	M4-M5 14PM	Experimental results for all scenarios
Activity 5 – Evaluation and reporting	KPI collection, user acceptance evaluation, analysis of results and documentation	M6 4PM	Final evaluation and reporting





Q&A
Thanks for your attention!

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